Theorem 4 The Principal Axes Theorem

Let $A \in M_{n\times n}$ be a symmetric matrix and

$A = PDP^T$ be an orthogonal diagonalization

i.e. $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$, P has orthonormal

column vectors $\mathcal{B} = \{u_1, \dots, u_n\}$ where $\mathcal{B}$ is a basis.

Then
$$Q(v) = v^T A v$$
$$= [v]_{\mathcal{B}}^T D [v]_{\mathcal{B}}.$$

Equivalently $Q(v) = y^T D y$ where

$$y = P^{-1} v \quad \text{or} \quad v = Py.$$

Example 5 in textbook.

Take $A = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$.

From $\det(A - xI) = (x-3)(x-7)$

we see that there are two distinct eigenvalues: 3 and 7.

$\mathrm{Nul}(A - 3I) = \dots = \mathrm{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$

$\mathrm{Nul}(A - 7I) = \dots = \mathrm{span}\left\{\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}$.

By Thm $\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\}$ is an orthogonal basis

Normalizing we obtain an orthonormal basis

$$B = \left\{ \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \right\} \text{ of } \mathbb{R}^2$$

of eigenvectors of A.

Let $P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$. Then

$A = P \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix} P^T$ is an orthogonal diagonalization.

Principal Axes Theorem tells us that

$$Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = [x \ \ y] \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= 5x^2 + 5y^2 - 4xy$$

is equal to $[x \ y]_B^T \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}_B$

Let's set $Q'\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = [x \ \ y] \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.

$$= 3x^2 + 7y^2$$

Let's compare Q with Q' and try to understand what the Principal Axes Thm is saying geometrically.

Before we start the comparison let's look at the quadratic form for the symmetric matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

It has quadratic form $Q''\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

$$= x^2 + y^2.$$

The quadratic form Q'' has axes of symmetry (Principal Axes) $\operatorname{span}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\operatorname{span}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The quadratic form Q' has axes of symmetry $\operatorname{span}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\operatorname{span}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

The quadratic form Q has axes of symmetry

$$\operatorname{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\} = \operatorname{span}\left\{\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}\right\} \text{ and}$$

$$\operatorname{span}\left\{\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\} = \operatorname{span}\left\{\begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}\right\}.$$

Observe that

$$Q'\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

$$= 3x^2 + 7y^2 > 0 \quad \text{for all } \begin{bmatrix} x \\ y \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ in } \mathbb{R}^2.$$

For this reason $Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x & y \end{bmatrix}_B \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}_B.$

$$> 0 \quad \text{for all } \begin{bmatrix} x \\ y \end{bmatrix}_B \neq 0$$

i.e. $Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) > 0$ for all $\begin{bmatrix} x \\ y \end{bmatrix} \neq 0$.

We'll see that Q is positive definite.

Example. Let $A = \begin{bmatrix} -5 & 2 \\ 2 & -5 \end{bmatrix} = -\begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$.

Then $A = P \begin{bmatrix} -3 & 0 \\ 0 & -7 \end{bmatrix} P^T$ with $P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$

and arguing from $-3x^2 + (-7)y^2 = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -3 & 0 \\ 0 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

we may conclude that $Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -5 & 2 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

$$= -5x^2 - 5y^2 + 4xy$$

$$< 0 \quad \forall \begin{bmatrix} x \\ y \end{bmatrix} \neq 0.$$

We call Q negative definite.

Example

Take $A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$.

It has orthogonal diagonalization

$$A = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}^T.$$

Let $Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

$$= x^2 - 5y^2 - 8xy$$

Let $Q'\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x & y \end{pmatrix} \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

$$= -7x^2 + 3y^2.$$

Clearly $Q'\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = -7(1)^2 + 3(0)^2 = -7 < 0$

$$Q'\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = 3 > 0.$$

So Q' is called indefinite.

Let $B = \left\{ \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}, \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} \right\}$ – an orthonormal basis for $\mathbb{R}^2$.

If $\begin{bmatrix} x \\ y \end{bmatrix}_B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ then by Principal Axes Thm

$$Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x & y \end{bmatrix}_B \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}_B$$

$$= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{bmatrix} 7 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= Q'\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = -7 < 0$$

Similarly if $\begin{bmatrix} x \\ y \end{bmatrix}_B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ then

$$Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \dots = Q'\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = 3 > 0.$$

So Q is also indefinite.

Example.

Take $A = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$. A has

orthogonal diagonalization

$$A = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}^T$$

Let $Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

$$= 3x^2 + 3y^2 + 6xy.$$

Is Q positive definite?

Is $Q'\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 6x^2$

positive definite? No, since $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

and $Q'\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = 6(0)^2 = 0$ which is <u>not</u> > 0.

We say Q' is indefinite.

If $B = \left\{ \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \right\}$ and $\begin{bmatrix} x \\ y \end{bmatrix}_B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

then $Q\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = Q''\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = 0$ so Q

is not positive definite either.

Def. Let $A \in M_{n\times n}$ be a symmetric matrix and $Q(v) = v^T A v$ be its quadratic form.

Then (a) Q is <u>positive definite</u> if $Q(v) > 0$ for all $v \neq 0$ in $\mathbb{R}^n$.

(b) Q is <u>negative definite</u> if $Q(v) < 0$ for all $v \neq 0$ in $\mathbb{R}^n$

(c) Q is <u>indefinite</u> if it is neither positive definite nor negative definite

Theorem 5

Let Q be the quadratic form of a symmetric $A \in M_{n\times n}$.

(a) Q positive definite $\iff$ all eigenvalues of A are positive (>0)

(b) Q negative definite $\iff$ all eigenvalues of A are negative (<0)

(c) Q indefinite $\iff$ some eigenvalues are non-negative and some eigenvalues are non-positive.

Example 6 in textbook.

Let $A = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{bmatrix} \in M_{3\times 3}$.

Then $Q\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = 3x_1^2 + 2x_2^2 + x_3^2 + 4x_1x_2 + 4x_2x_3$

We can tell whether Q is positive/negative definite or indefinite just by look at its eigenvalues.

$$\det(A - xI) = \cdots = -(x+1)(x-2)(x-5) = 0$$

$$\Leftrightarrow \quad x = -1, 2, 5$$

By Thm 5 Q is indefinite.